

9.4.35

EE25BTECH11022 - sankeerthan

Question: A train travels a distance of 480km at a uniform speed. If the speed had been $8\frac{\text{km}}{\text{h}}$ less, then it would have taken 3 hours more to cover the same distance. We need to find the speed of the train.

Solution: Let the unknown speed be x .

$$t = 480 \left(\frac{1}{\frac{1}{x-8}} \right) \quad (1)$$

$$\mathbf{A}^T \mathbf{X} = 3, \text{ where } \mathbf{A} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} \frac{480}{x-8} \\ \frac{480}{x} \end{pmatrix} \quad (2)$$

Multiply both sides by $x(x-8)$:

$$480x - 480(x-8) = 3x(x-8) \quad (3)$$

$$480x - 480x + 3840 = 3x^2 - 24x \quad (4)$$

$$3x^2 - 24x - 3840 = 0 \quad (5)$$

$$\begin{pmatrix} x^2 & x & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -24 \\ -3840 \end{pmatrix} = 0 \quad (6)$$

Discriminant:

$$D = (-24)^2 - 4 \times 3 \times (-3840) = 46656 \quad (7)$$

$$x = \frac{24 \pm \sqrt{46656}}{6} = \frac{24 \pm 216}{6} \quad (8)$$

Positive root:

$$x = \frac{24 + 216}{6} = 40 \quad (9)$$

Therefore, train speed = $40\frac{\text{km}}{\text{h}}$